

CIS 6270

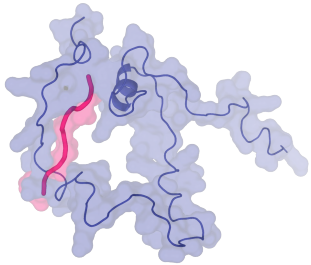
Module 5: Discrete Flow Matching

Lecture 5.5: Rectification

Tue, Nov 10, 2026

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Rectified Flow

FLOW STRAIGHT AND FAST: LEARNING TO GENERATE AND TRANSFER DATA WITH RECTIFIED FLOW

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Liu,..,Liu. *ICLR* (2023)

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We need many small steps to follow a curved trajectory. A straight trajectory can be traversed in one step.

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We call the repeated procedure reflow. The authors report that errors accumulate after a few rounds.

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By the triangle inequality, the straight segment between the rewired endpoints is no longer than the path that was rewired.

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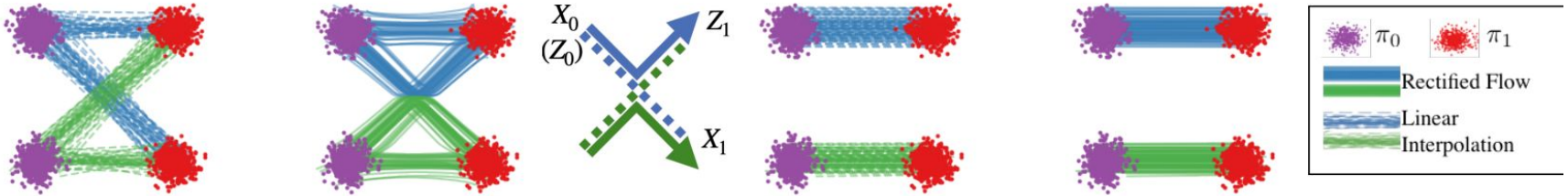
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This illustrates the benefit of a better coupling. Rectification uses generated paths to obtain new training pairs.

After reflow, the new endpoint pairs follow straight paths in this example



(a) Linear interpolation $X_t = tX_1 + (1-t)X_0$

(b) Rectified flow Z_t induced by (X_0, X_1)

(c) Linear interpolation $Z_t = tZ_1 + (1-t)Z_0$

(d) Rectified flow Z'_t induced by (Z_0, Z_1)

(a) Independent pairs cross. (b) The flow rewires them. (c, d) The new pairing is straight.

Liu,..,Liu. *ICLR* (2023)

Could a better pairing also preserve dependencies between tokens in one parallel step?

ReDi: Rectified Discrete Flow

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Yoo,..,Hong. *NeurIPS* (2025)

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In one discrete step, we draw every position using only the current sequence. The draws do not condition on each other.

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With one large step, we set all positions at once and lose the dependencies between their new tokens.

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It is a property of the coupling. The same data distribution can have total correlation zero or high, depending on how pairs are drawn.

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A large value means parallel token predictions miss combinations that the true sequence distribution requires.

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One parallel step from this coupling produces a complementary pair one time in four.

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We call this procedure ReDi. We use the generated pairs to train the next round.

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The result is a non-increasing sequence bounded below by zero, so the error converges, though not necessarily to zero.

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The limiting error is positive because some source states are equally close to two valid sequences. The coupling assigns probability to both.

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Independent position draws also produce AC and GT. Half the outputs miss the two-sequence data support.

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Toy data are AT and GC, each with probability 0.5. There are two equally likely source states. With independent pairing, either source gives both endpoints probability 0.5.

$$TC = 2\left[\tfrac{1}{2}\log\frac{1/2}{1/4}\right] = \log 2 \approx 0.693, \qquad p(\text{valid parallel draw}) = \tfrac{1}{2}$$

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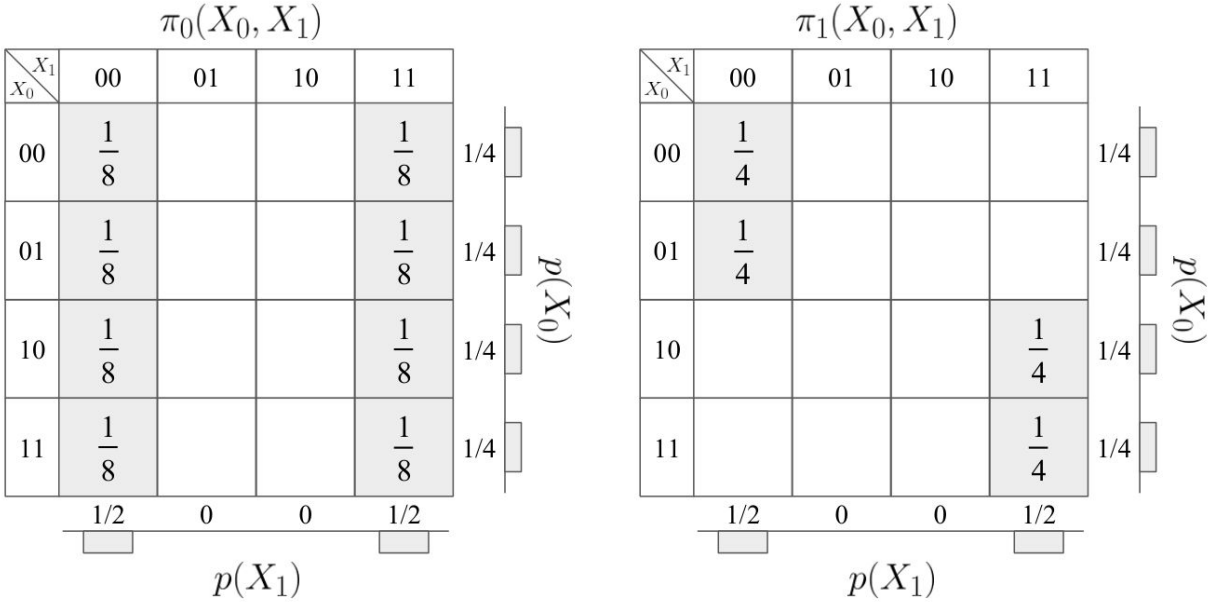
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If a learned coupling assigns one endpoint to each source, a parallel draw becomes exact in this toy case.

A more informative source can remove the dependence that the output positions must still coordinate.

The couplings have the same marginals, but only the deterministic pairing supports exact one-step generation



Left: independent pairing spreads each source over both targets. Right: a deterministic pairing.

Yoo,..,Hong. *NeurIPS* (2025)

One-step generation on ImageNet

Step	Model	FID (↓)	IS (↑)	Prec. (↑)	Rec. (↑)	Den. (↑)	Cov. (↑)
1	MaskGIT [3]	95.16	12	0.26	0.12	0.17	0.35
	SDTT† [10]	90.40	14	0.31	0.13	0.21	0.34
	Di4C [17]	90.32	13	0.26	0.24	0.17	0.33
	ReDi[1]	37.43	49	0.63	0.51	0.78	0.86
	ReDi[2]	21.80	90	0.74	**0.52**	1.05	0.93
	ReDi[3]-distill	**11.68**	**182**	**0.83**	0.44	**1.25**	**0.96**
4	MaskGIT [3]	10.90	184	0.83	0.46	1.18	0.96
	SDTT† [10]	8.97	205	**0.88**	0.41	**1.43**	0.97
	Di4C [17]	**6.20**	216	0.87	**0.52**	1.33	**0.98**
	ReDi[1]	7.58	228	0.87	0.46	1.33	**0.98**
	ReDi[2]	7.86	**240**	0.87	0.44	1.31	0.97
8	MaskGIT [3]	6.51	227	0.89	0.48	1.38	0.98

Yoo,..,Hong. *NeurIPS* (2025)

One-step generation with the masked model has FID 95

Step	Model	FID (↓)	IS (↑)	Prec. (↑)	Rec. (↑)	Den. (↑)	Cov. (↑)
	MaskGIT [3]	95.16	12	0.26	0.12	0.17	0.35
1	SDTT† [10]	90.40	14	0.31	0.13	0.21	0.34
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Yoo,..,Hong. *NeurIPS* (2025)

After one round of rectification, the same architecture reaches 37.4

Step	Model	FID (↓)	IS (↑)	Prec. (↑)	Rec. (↑)	Den. (↑)	Cov. (↑)
1	MaskGIT [3]	95.16	12	0.26	0.12	0.17	0.35
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Yoo,..,Hong. *NeurIPS* (2025)

After two rounds, 21.8

Step	Model	FID (↓)	IS (↑)	Prec. (↑)	Rec. (↑)	Den. (↑)	Cov. (↑)
1	MaskGIT [3]	95.16	12	0.26	0.12	0.17	0.35
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Yoo,..,Hong. *NeurIPS* (2025)

Distilling the third coupling into a one-step model reaches 11.7

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1	MaskGIT [3]	95.16	12	0.26	0.12	0.17	0.35
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Yoo,..,Hong. *NeurIPS* (2025)

The four-step masked model reaches 10.9, so one step now nearly matches four

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Yoo,..,Hong. *NeurIPS* (2025)

Once generation is fast, could we use the saved computation to improve several properties?

AReUReDi: Annealed Rectified Updates for Refining Discrete Flows

AReUReDi: Annealed Rectified Updates for Refining Discrete Flows with Multi-Objective Guidance

Tong Chen,[1] **Yinuo Zhang,**[2] **Pranam Chatterjee**[1,3,†]

[1]Department of Computer and Information Science, University of Pennsylvania
[2]Centre for Computational Biology, Duke-NUS Medical School, Singapore
[3]Department of Bioengineering, University of Pennsylvania

[†]Corresponding author: pranam@seas.upenn.edu

Chen,..,Chatterjee. *arXiv* (2025)

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With guidance we steered the rates and kept the sampler. In AReUReDi we rectify the model first, then refine each sample.

Specifically, we rectify the peptide flow three times, then use its per-position marginals as the refinement proposal prior.

In AReUReDi, we refine a sequence one position at a time

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Score a sequence by its weakest weighted objective:

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Increasing an objective's weight increases its weighted value. It is then less likely to be the minimum, so optimization places less pressure on it.

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$$\eta_t = \eta_{\min} + (\eta_{\max} - \eta_{\min})\,\frac{t}{T-1}$$

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We explore broadly at 1 and favor high-scoring sequences at 20.

Propose a new letter at one position, reweighted by the tilt

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Propose a new letter y at position i from the model, reweighted by the tilt ratio:

$$q_i(y \mid x) = \frac{p_t^i(y \mid x)\; g\big(r_i(y;x)\big)}{Z(x)}, \qquad r_i(y;x) = \frac{W_{\eta,\omega}\big(x^{(i \leftarrow y)}\big)}{W_{\eta,\omega}(x)}$$

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The proposal $q_i(y \mid x)$ is the model's own distribution p^i_t over letters at position i, times a balancing function g of the ratio r_i between the tilt at the sequence with y placed at position i and the tilt at the current sequence, divided by the normalizer Z(x) that sums that product over the candidate letters.

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The balancing function satisfies g(u) = u g(1/u). Barker's choice is u/(1 + u) and the square root also works.

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By that identity, the proposal ratio has one factor of the target ratio, as the acceptance step requires.

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$$\alpha_i(x, x') = \min\left\{1,\ \frac{\pi_{\eta,\omega}(x')\, q_i(x^i \mid x')}{\pi_{\eta,\omega}(x)\, q_i(y \mid x)}\right\}, \qquad x' = x^{(i \leftarrow y)}$$

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With the acceptance step in place the chain satisfies detailed balance, so the refinement has the tilted distribution as its stationary law.

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A move that is easy to propose may need to be rejected so it does not receive too much probability.

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The paper states that acceptance is always one. It is one when the two normalizers agree. In the paper’s example, their ratio is 1.38 in one direction, giving acceptance probability 0.72 for the reverse move.

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As η grows the right side goes to zero, so the samples concentrate on the maximizers of the weighted worst objective.

Every maximizer is weakly Pareto optimal. A unique maximizer is Pareto optimal.

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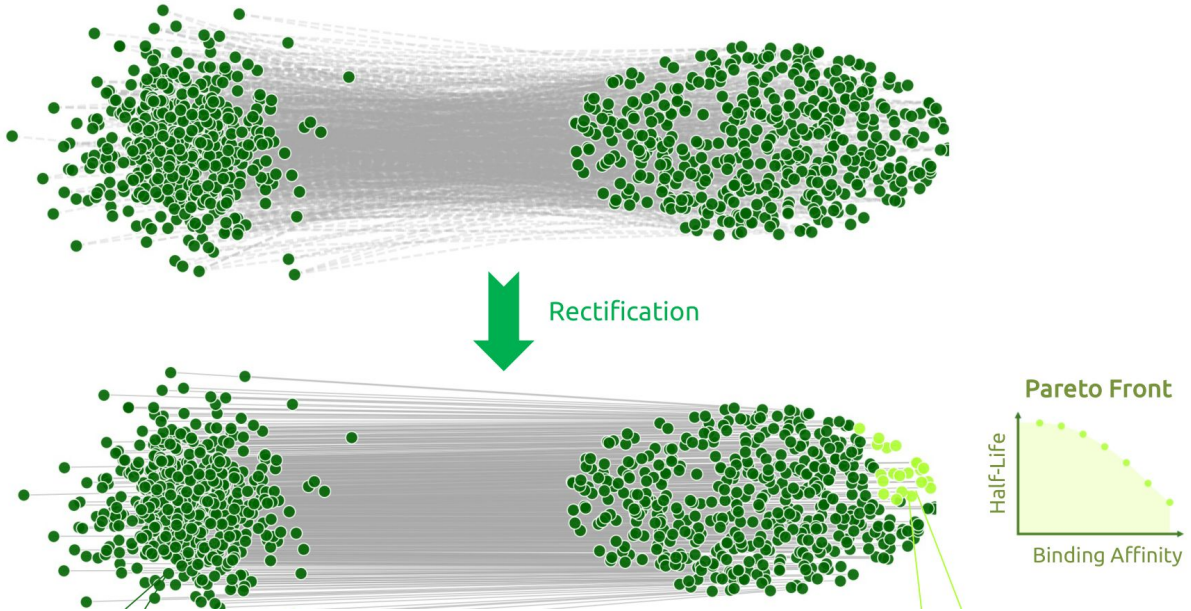
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The acceptance correction keeps the tilted distribution consistent with the proposal mechanism.

In AReUReDi we rectify first, then take a Metropolis step at one position per iteration



We rectify the coupling to make paths parallel before applying guidance.

Chen,..,Chatterjee. *ICLR Workshop* (2026)

AreUReDi performs stronger than all multi-objective baselines

Target	Method	Time (s)	Hemolysis	Non-Fouling	Solubility	Half-Life (h)	Affinity
1B8Q	MOPSO	8.54	0.8934	0.4763	0.4684	4.45	6.0594
	NSGA-III	33.13	0.9138	0.5715	0.5825	7.32	7.2178
	SMS-EMOA	8.21	0.8804	0.3450	0.3511	3.02	5.955
	SPEA2	17.48	0.9181	0.4973	0.5057	4.13	**7.3240**
	PepTune + DPLM	**2.46**	0.8547	0.3085	0.3213	1.17	5.2398
	AReUReDi	55	**0.9214**	**0.8680**	**0.8654**	**22.93**	5.7130
PPP5	MOPSO	11.34	0.9117	0.4711	0.4255	1.77	6.6958
	NSGA-III	37.30	**0.9521**	0.7138	0.7066	2.90	7.3789
	SMS-EMOA	8.43	0.8758	0.4269	0.4334	1.03	6.2854
	SPEA2	19.02	0.9445	0.6221	0.6098	2.61	**7.6253**
	PepTune + DPLM	**4.80**	0.8816	0.2752	0.2636	1.27	5.8454
	AReUReDi	195	0.9412	**0.896**	**0.8832**	**38.28**	6.7186

Chen,..,Chatterjee. *arXiv* (2026)

AreUReDi balances multiple properties when generating molecules

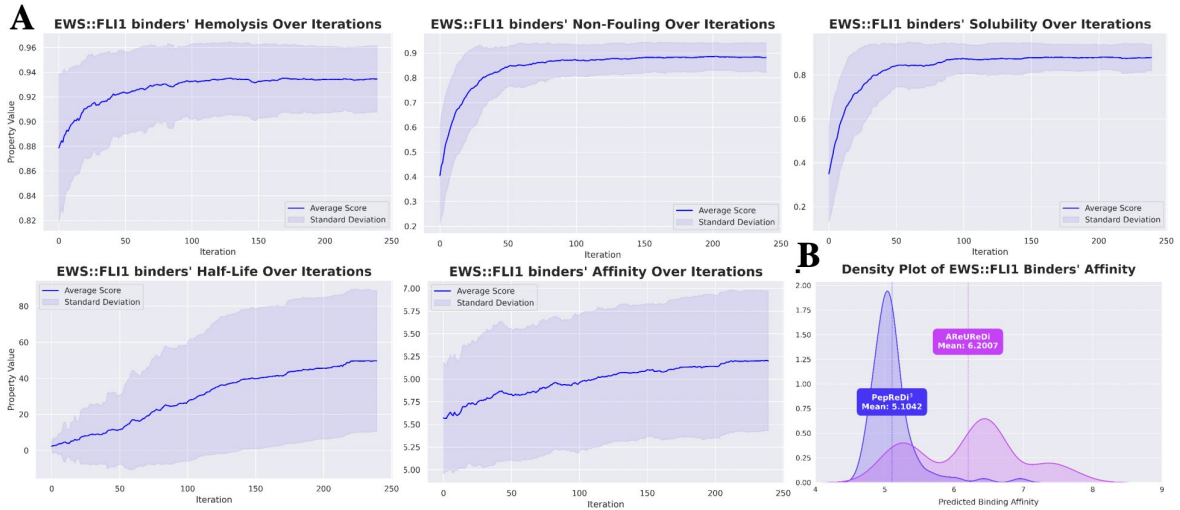


Figure 2: **Guided peptide refinement for EWS::FLI1.** Panels A and B show predicted-property trajectories with standard-deviation bands across 100 length-12 candidates and the affinity distribution of guided versus unconditional PepReDi[3] samples, respectively. The hemolysis score is oriented toward non-hemolysis.

Chen,..,Chatterjee. *arXiv* (2026)

Let’s now review!

State spaces, paths, and velocities

Choose a path, write its velocity, average over endpoints

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Sampling is an Euler step, with a corrector, a guidance tilt, or a Metropolis step added on top.

Let’s put the choices side by side

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Edit Flows: sequences of any length, an alignment, insertion, deletion, and substitution rates.

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Rectification: a coupling with little structure left between positions, so one parallel step can work.

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Discrete flow matching: $$u_t^i(\cdot, x) = \frac{\dot{\kappa}_t}{1 - \kappa_t}\left[p_{1|t}(\cdot \mid x) - \delta_{x^i}\right]$$

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On the simplex, the velocity points toward the data corner at a Beta-function speed. On the sphere, it is the logarithm map toward the data divided by the remaining time.

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Guidance multiplies each rate by the ratio of the expected terminal reward at the destination to the one at the current state.

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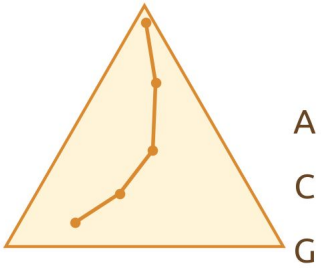
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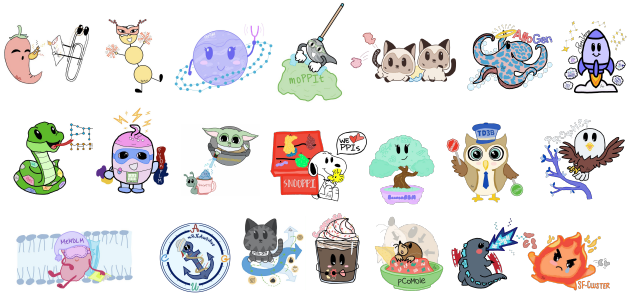
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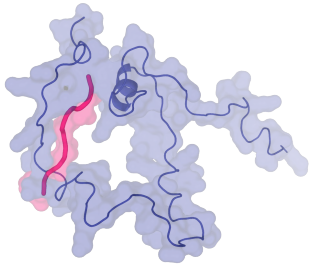
Rectifying the coupling does not raise the factorization error of a one-step sampler.

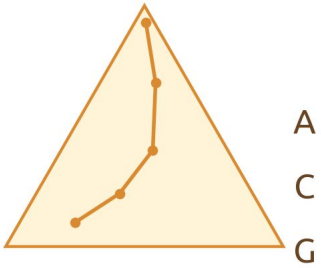


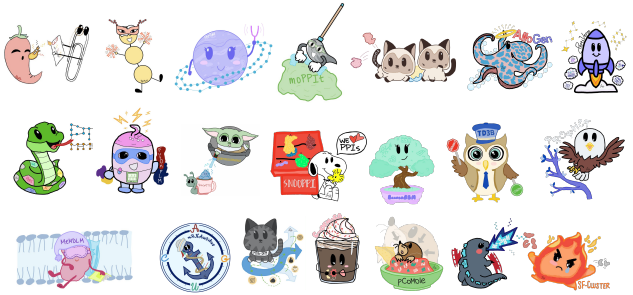


Can we learn the destination directly?

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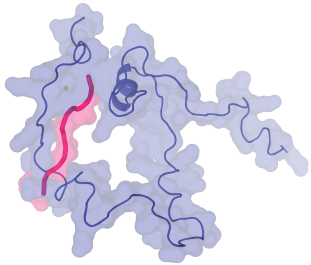


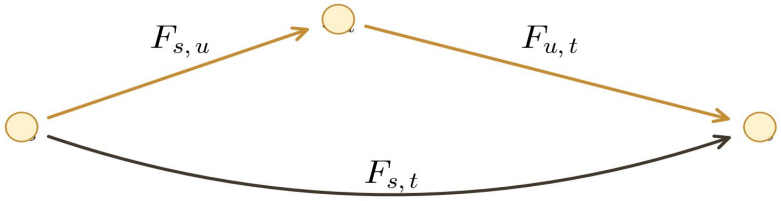


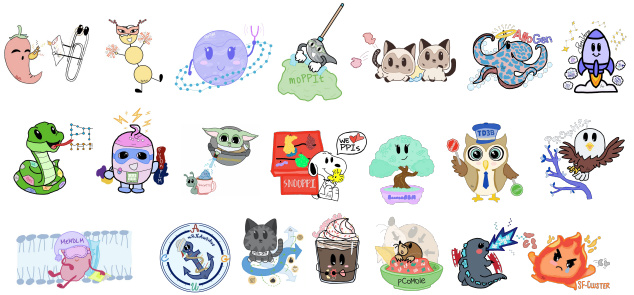


Can we learn the destination directly?
That leads us to Flow maps!

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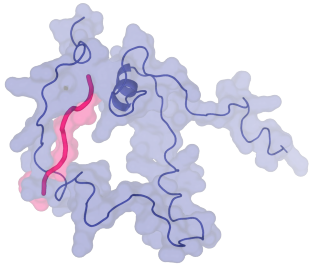
Module 6: Flow Maps

Lecture 6.1: Flow Maps and the Three Training Identities

Tue, Nov 10, 2026

Pranam Chatterjee, PhD
Assistant Professor of Bioengineering
Assistant Professor of Computer and Information Science
University of Pennsylvania

Penn Engineering
University of Pennsylvania



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Each application costs one network evaluation, so a sample costs as many evaluations as we take steps.

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The solution at t = 1 is the generated sample. Can we predict this endpoint with only a few evaluations?

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At t = 1 the exact answer is e, which is 2.71828.

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In one Euler step we add the step size one times the velocity at the starting point, which is one, and get two.

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Now solve the same equation with Euler's method

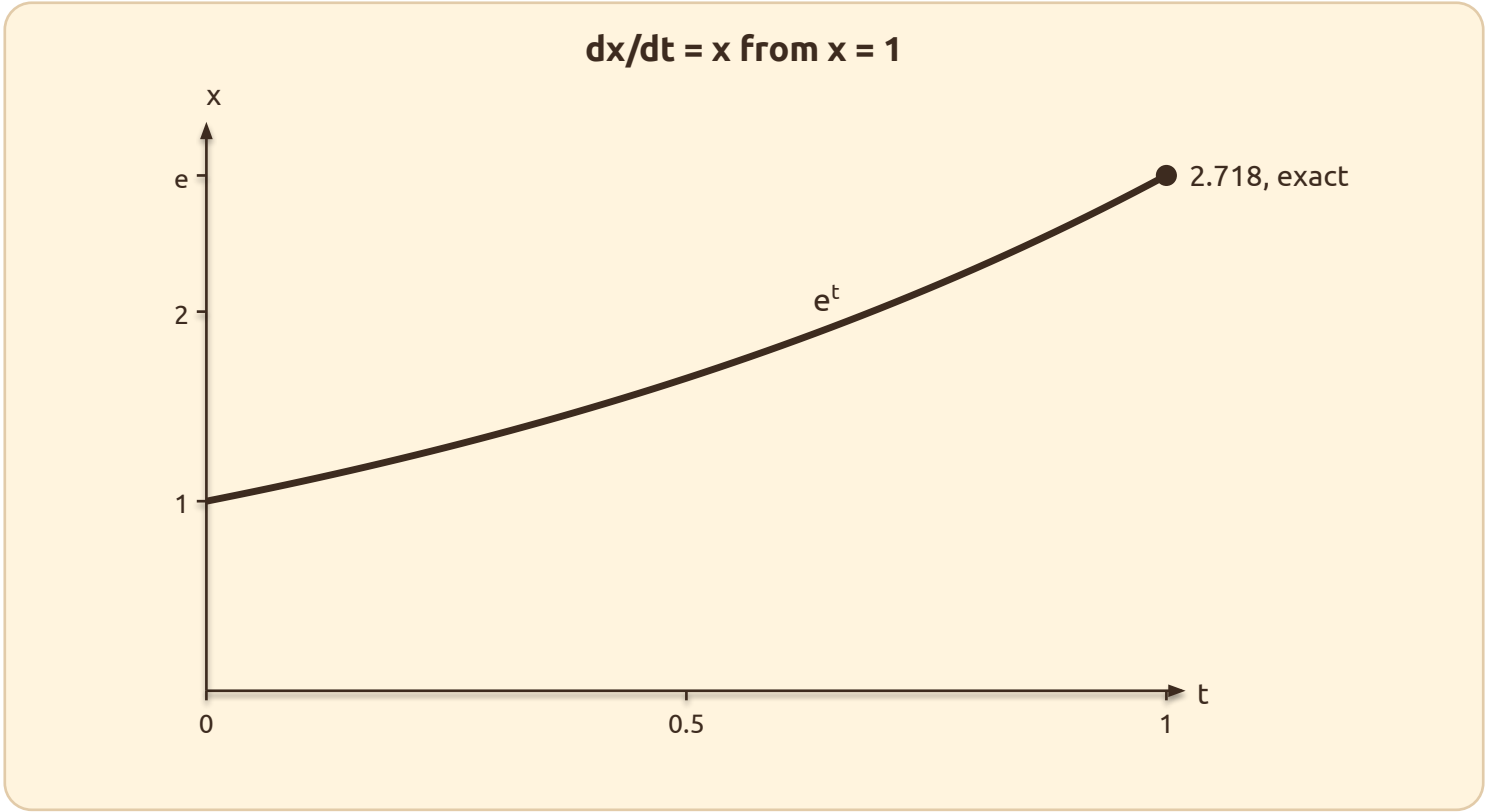
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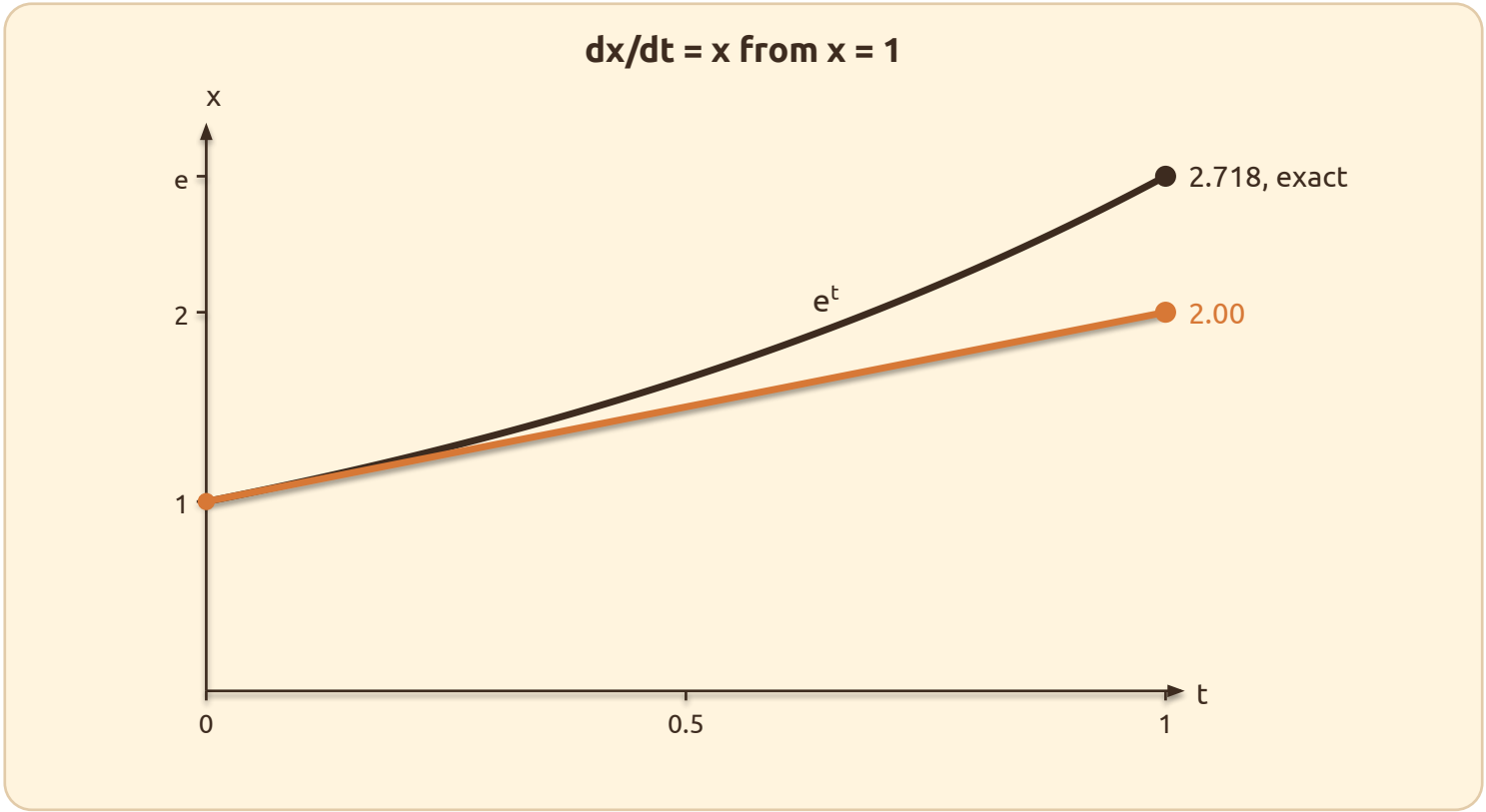
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The first half step reaches 1.5. In the second step, we add half the updated velocity, 1.5, to reach 2.25.

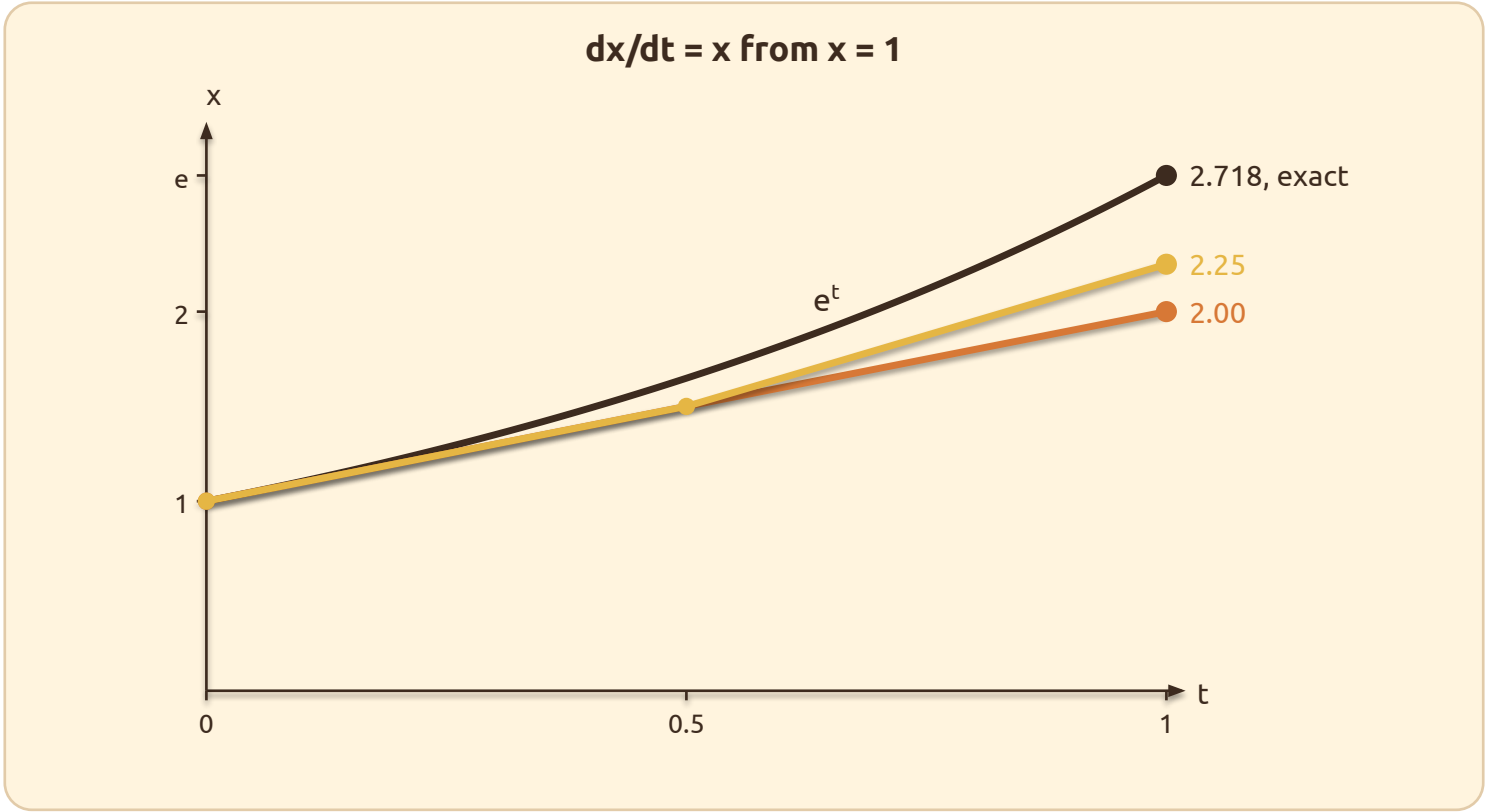
Here is the exact solution from t = 0 to t = 1



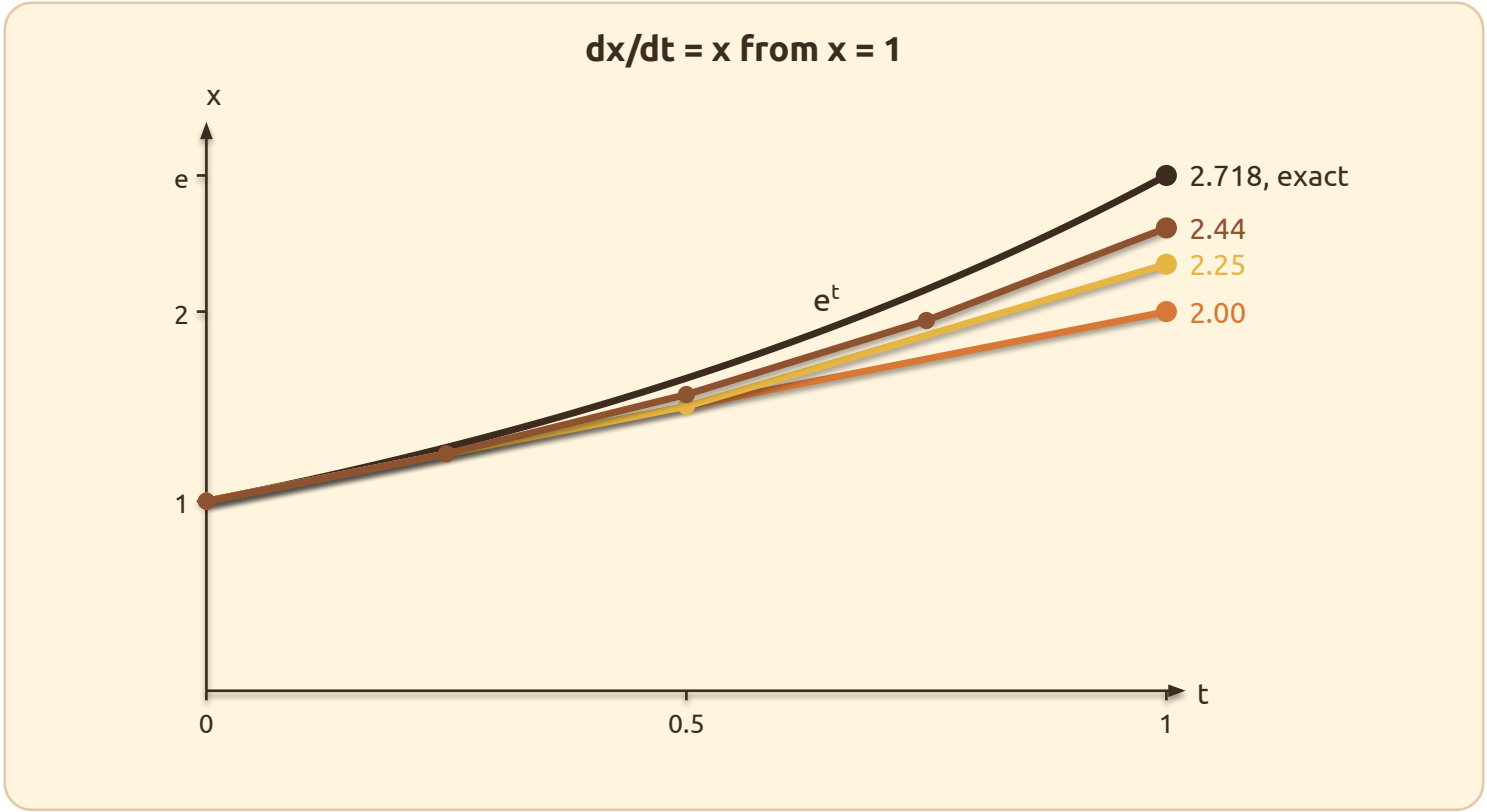
One step of size one lands at 2



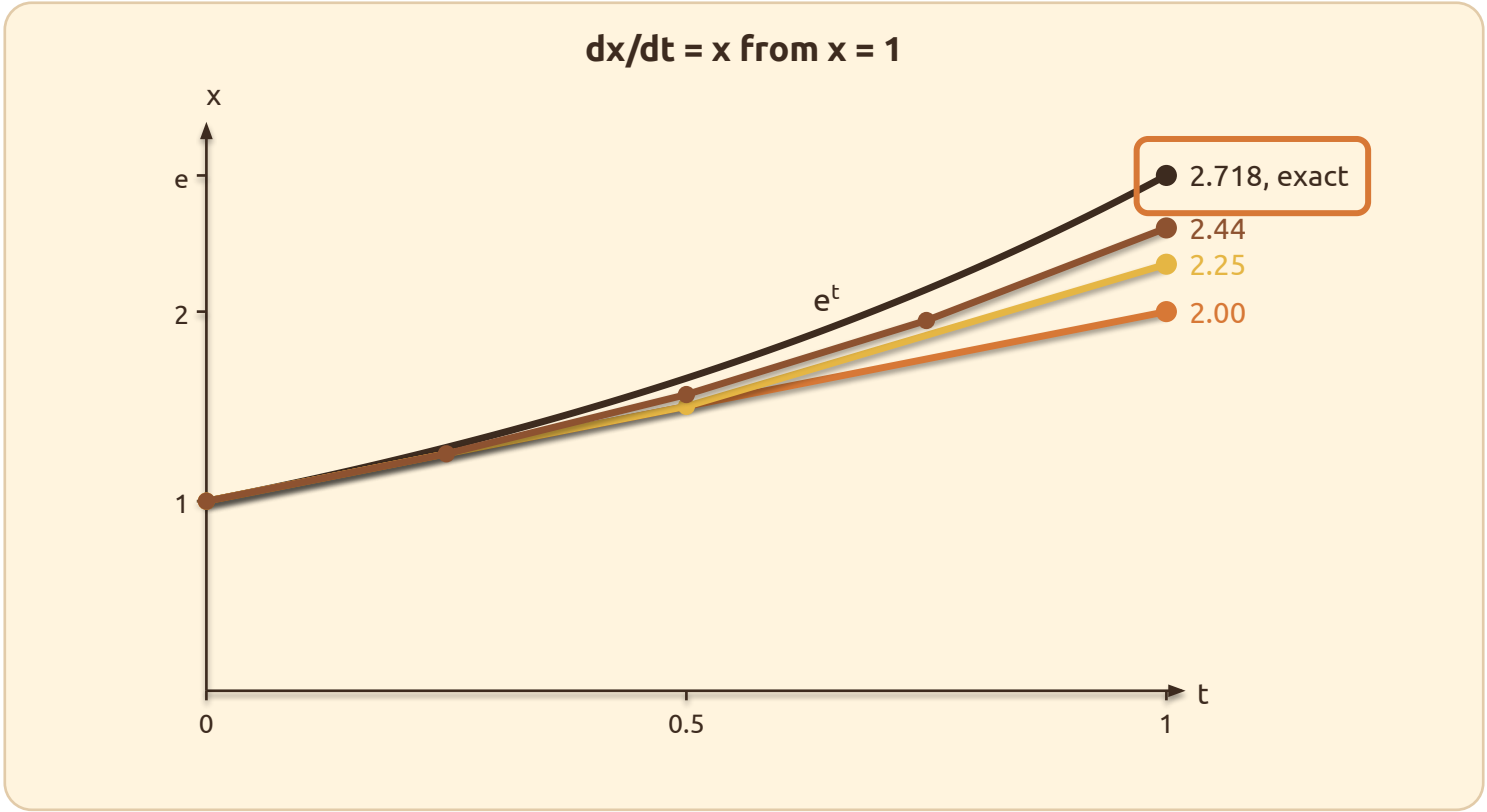
Two steps reach 2.25



Four steps reach 2.441



Euler’s endpoint approaches e at rate 1/N



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Each of those N steps is one network evaluation in a real model.

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Before we learn this function, let's define it as the solution of the differential equation.

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We can bound how velocity errors grow along a trajectory using the constant L.

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A generative sample is $X_{0,1}(x_0)$, one evaluation of this map.

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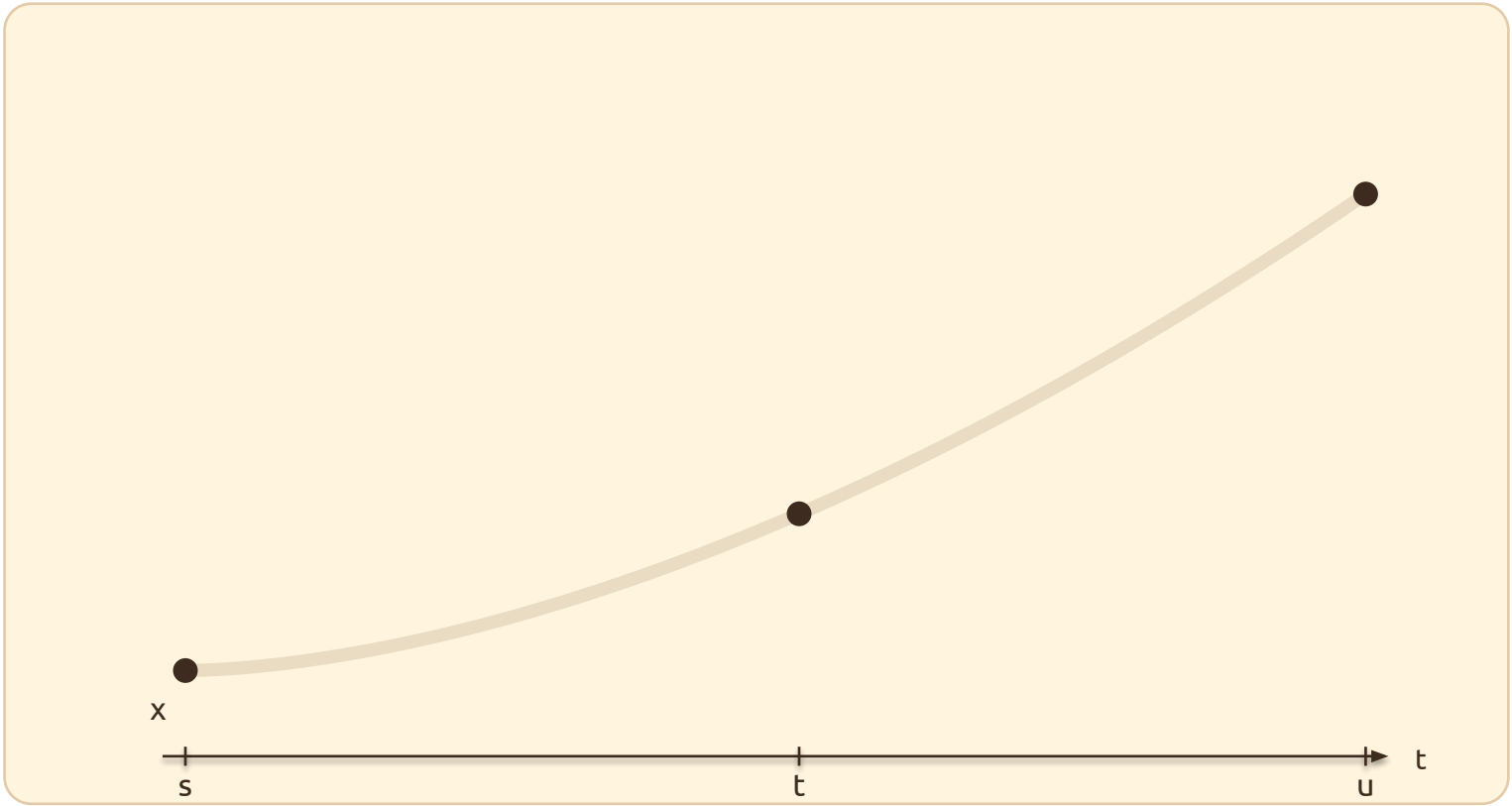
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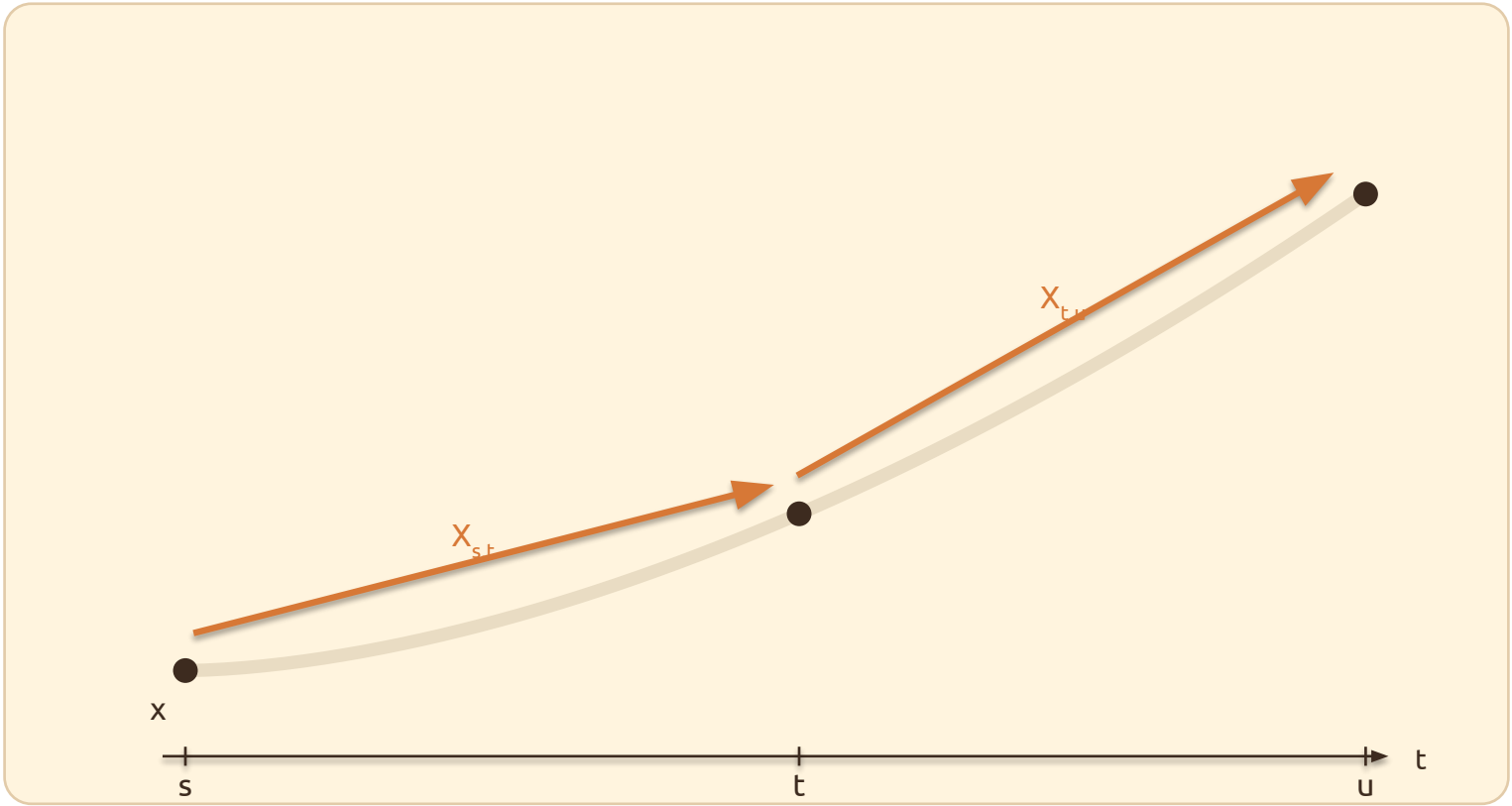
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A model satisfies it automatically if we write the map as x plus (t – s) times a network output.

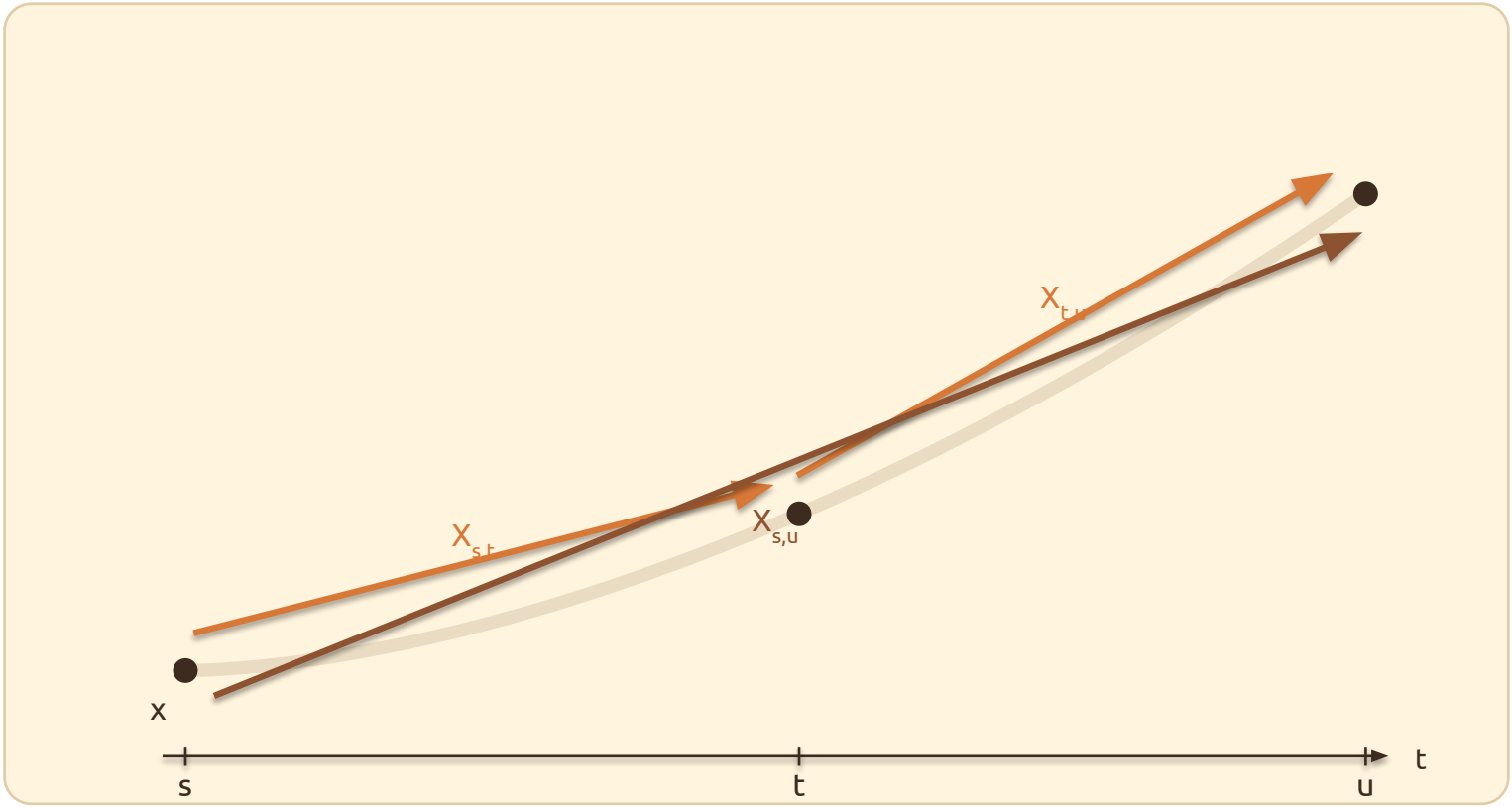
Next, draw the cocycle axiom on one trajectory



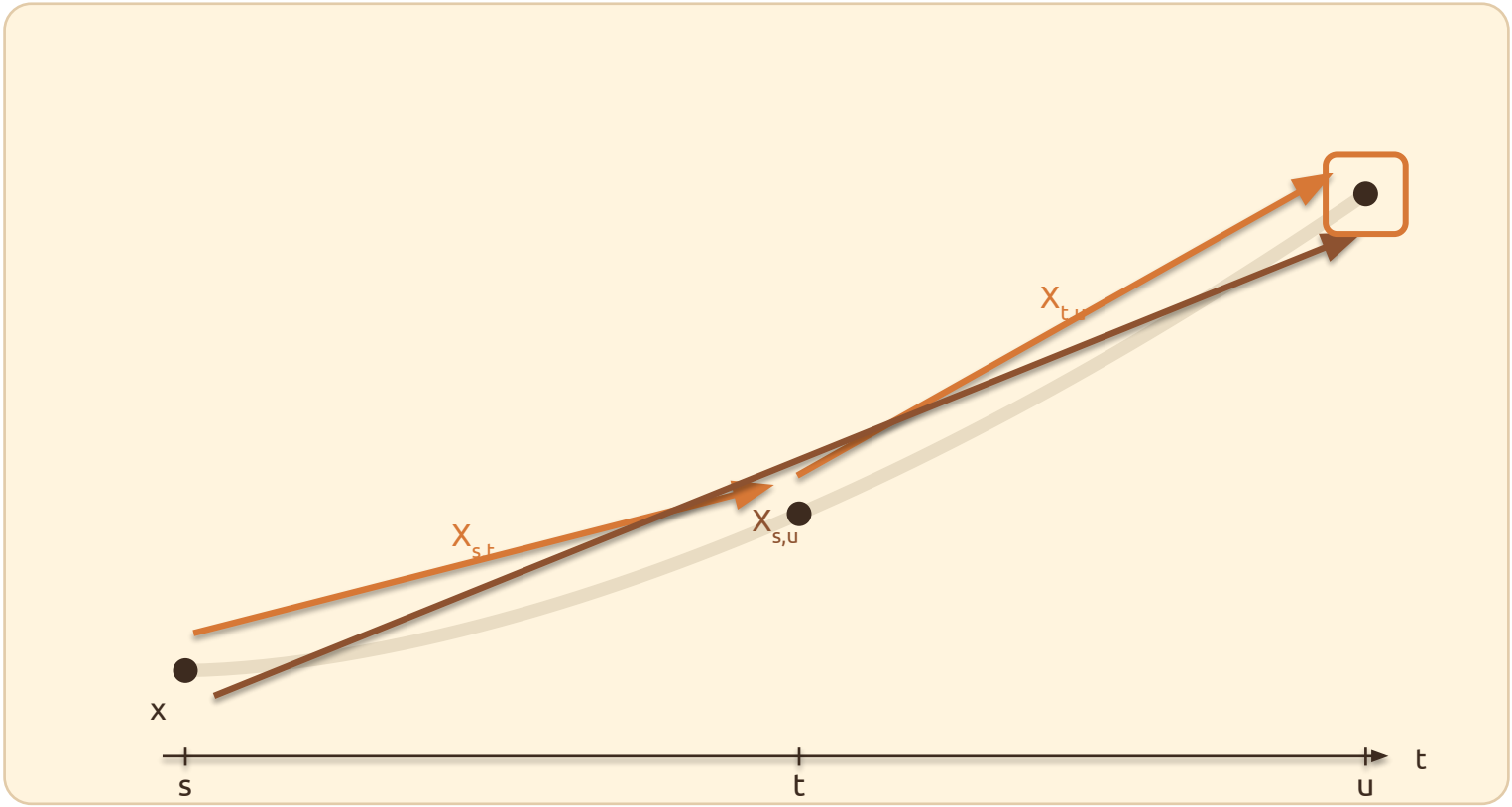
Follow the trajectory from s to t, then from t to u



Follow the same trajectory directly from s to u



Both land on the same point of the same trajectory



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Few-step sampling comes from this identity, since one long move equals several short ones.

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Without this axiom, the first two do not determine the map. A model can satisfy both and still leave every sample unchanged.

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If the velocity does not depend on t, only the elapsed time matters:

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When only forward time is available, which happens on discrete states and for stochastic paths, the group becomes a semigroup.

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So every flow map is a bijection of the state space onto itself.

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We can compute the volume change and use it to evaluate exact likelihoods.

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It measures how the destination moves when the starting point moves a little.